

TravelLink — Smart Itinerary Builder

Progress Notes

Last updated: 2026-05-28

✅ Fully Completed Modules

- Auth, Friends, Groups, Image Upload (Cloudinary), User, Expense, Location (SignalR)
 - All Itinerary entities + ApplicationDbContext + migration (`AddedItineraryModule`)
 - All Itinerary DTOs
 - `IItnineraryService.cs` — 10 method signatures
 - `ItineraryService.cs` — all 10 methods implemented
 - `ItineraryController.cs` — all 10 endpoints
 - `Program.cs` — `AddHttpClient()` + `AddScoped<IItnineraryService, ItineraryService>()` registered
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✅ Bugs Fixed This Session

1. 401 on `/api/itinerary/{requestId}/generate`

- **Cause:** Missing/expired JWT in Swagger — not a code bug
- **Fix:** Re-authenticate via `POST /api/auth/login` → click  Authorize in Swagger

2. Gemini JSON parse crash (backtick fence)

- **Cause:** Gemini 2.5 Flash wraps response in ````json ... ```` despite prompt
- **Fix:** Strip markdown fences before `JsonDocument.Parse` (added in `GenerateItineraryAsync` ~line 253)

3. Invalid column name 'Date'

- **Cause:** `ItineraryDay.Date` was added to entity after migration ran — DB table missing the column
- **Fix:** `dotnet ef migrations add FixItineraryDayAndItemSchema` + `dotnet ef database update`

4. Duplicate key on `GeneratedItineraries`

- **Cause:** Previous failed runs leave orphaned `GeneratedItinerary` rows
- **Fix:** Added orphan cleanup before inserting new `GeneratedItinerary` in `GenerateItineraryAsync`

5. `dropped_suggestions` `NullReferenceException`

- **Cause:** Gemini sometimes omits `dropped_suggestions` key entirely
- **Fix:** Changed `GetProperty` → `TryGetProperty`, store as nullable `droppedJson`

6. `TotalDays = 0` in response

- **Cause:** `TotalDays` never set on `GeneratedItinerary` entity when saving
- **Fix:** Added `TotalDays = totalDays` to `GeneratedItinerary` creation ✅ (user fixed)

7. `BudgetPerPersonPerDay = 0` in response

- **Cause:** Not mapped in `GetItineraryResultAsync` result construction
- **Fix:** Added `BudgetPerPersonPerDay = request.BudgetPerPersonPerDay + RequestId + TotalDays` mappings ✅ (user fixed)

🚧 Pending Improvements (Deferred)

Prompt Improvement 1 — Budget clarity

- Current prompt: `Budget:{request.BudgetPerPersonPerDay}/person/per` (truncated, model misreads it)
- Fix: `Budget: INR {request.BudgetPerPersonPerDay} per person per day – TOTAL daily allowance covering ALL meals, activities, and transport combined`

Prompt Improvement 2 — Scooty rental throughout trip

Plan (two-step Gemini approach):

1. **Pre-fetch call** to Gemini: ask for scooty availability + daily rate at the destination → returns JSON `{ available, daily_rate_per_scooty_inr, popular_rental_areas, notes }`
2. **Inject rental context** into main itinerary prompt
3. **Transport rules** in prompt:
 - Scooty = default transport ALL days (1 per 2 group members)
 - Day 1: include scooty pickup as first item (full-trip rental cost)

- Only use cab for: airport/railway transfers, distances >50km, bad weather
- Scooty cost per person per day auto-calculated from pre-fetch data

Prompt Improvement 3 — Group size context

- Fetch `groupMemberCount` from DB → include in prompt
- `scootyCount = ceil(groupMemberCount / 2.0)`

Current API Endpoints (all working

Method	Route	Description
POST	<code>/api/itinerary/request</code>	Create itinerary request (admin only)
POST	<code>/api/itinerary/{requestId}/suggest</code>	Add place suggestion
DELETE	<code>/api/itinerary/suggestion/{id}</code>	Delete suggestion
POST	<code>/api/itinerary/suggestion/{id}/vote</code>	Toggle vote on suggestion
PATCH	<code>/api/itinerary/suggestion/{id}/review</code>	Admin approve/reject suggestion
GET	<code>/api/itinerary/{requestId}/suggestions</code>	List all suggestions
POST	<code>/api/itinerary/{requestId}/generate</code>	Generate itinerary via Gemini
GET	<code>/api/itinerary/{requestId}/result</code>	Get generated itinerary
PATCH	<code>/api/itinerary/item/{itemId}/complete</code>	Toggle item complete
POST	<code>/api/itinerary/{requestId}/regenerate</code>	Regenerate (deletes old, makes new)

Config Notes

- Gemini model: `gemini-2.5-flash` in `appsettings.json` → `Gemini:Model`
- Gemini API URL:
`https://generativelanguage.googleapis.com/v1/models/{model}:generateContent?key={apiKey}`
- Weather: Open-Meteo (free, no key needed)

- AI Studio key required (not Google Cloud Console key) for free tier access
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Next Steps (when resuming)

1. Implement the two-step scooty rental pre-fetch (detailed plan above)
2. Fix budget wording in prompt
3. Add group size context to prompt
4. Consider Angular frontend integration